COMP20008 Assignment 2 — Sample Report Marking

Criterion	Max	Awarded	Level
Methods	**7**	**5.32**	
Data Preprocessing	1	0.66	Good
Correlation Analysis	1	0.66	Good
Supervised Learning Models	2.5	2	Very good
Feature Selection	0.5	0.5	Excellent
PCA	0.5	0.5	Excellent
Clustering	1.5	1	Good
Data Analysis & Interpretation	**15**	**7.75**	
Impact of Preprocessing on Results	1.5	1	Good
Correlation Interpretation	2.5	1.25	Good
Model and Evaluation Comparison	2.5	1.25	Good
Feature Selection Interpretation	2	1	Good
Limitations and Improvement Opportunities	1.5	0.5	Basic
PCA Interpretation	1.5	1	Good
Clustering Interpretation	2.5	1.25	Good
Cross-Section Consistency	1	0.5	Basic
Report Quality	**2**	**2**	
Visualisations	1	1	Good
Formatting & Presentation	1	1	Good
Total (scaled to 18)	**24**	**15.07 (11.30)**	

1. Methods

1.1. Data Preprocessing — 0.66 / 1 (Good)

What earned marks. Six distinct candidates, each tied to a computed property of the data (Table 1 of the report): one duplicate pair (ids 102 and 143), r = 0.963 between petal length and width, four IQR outliers named by id. All three steps are implemented correctly — the duplicate is removed before splitting, and the scaler is fitted inside the CV pipeline so no validation data leaks into it.

What held it back. The case for standardisation is a textbook sentence ("features with larger values would dominate the distance calculation") that could appear in any report. The group computed the standard deviations but never asked which feature would dominate and whether that matters for this RQ. The closing link to the RQ ("cleaner and more comparable inputs") is generic.

What an Excellent response would say here

"Unscaled, petal_length (SD 1.768) contributes about four times the spread of sepal_width (SD 0.436) to Euclidean distance. Because our RQ asks which measurements distinguish species, scaling is not neutral here: it raises the weight of sepal_width, the variable least associated with species, to equal that of the petal measurements. We apply it so that KNN tests all four measurements on equal terms, and we report its effect on KNN in Section 3.1."

1.2. Correlation Analysis — 0.66 / 1 (Good)

What earned marks. The variable set is justified against the RQ, the four species pairs are correctly marked N/A for Pearson and Spearman with a reason (nominal target, arbitrary coding), and all 40 values are computed correctly and reproducibly.

What held it back. The discretisation choice (5 equal-frequency bins) is stated but not justified. MI computed on binned data depends on the number of bins, so this is an implementation choice the rubric asks to be justified against the dataset.

> **What an Excellent response would say here**
>
> *"With 149 rows, 5 equal-frequency bins give about 30 flowers per bin, so each cell of a 5 × 5 contingency table expects about 6 flowers. With 10 bins the expected count drops to about 1.5 per cell, and MI is biased upwards by sparse cells. We therefore used 5 bins."*

1.3. Supervised Learning Models — 2 / 2.5 (Very good)

What earned marks. Clear target; split justified with the actual class counts (40/40/39 training, 10/10/10 test); 5-fold stratified CV justified with fold sizes (about 24 rows, about 8 per class); all 32 configurations reported (Table 4); every changed hyperparameter given as default → chosen → effect (Table 5); per-class precision, recall and F1 for both models and the baseline.

What held it back. The metric justification has one weak link. Macro-F1 is chosen because "all three species are equally important", but with classes of 10/10/10 in the test set, macro-F1 and accuracy coincide (both 0.967 for KNN), so the choice adds nothing the report notices. The selected k = 3 is also justified only by its mean CV score (0.964 vs 0.946 for k = 5), although the fold-to-fold standard deviation at k = 3 is 0.053, larger than the 0.018 gain.

> **What an Excellent response would say here**
>
> *"Because the classes are balanced, macro-F1 equals accuracy (0.967 for KNN) and cannot separate the models any further. The metric that answers our RQ is per-class recall: it shows that every error falls between versicolor and virginica (versicolor recall 0.800 for the tree, virginica recall 0.900 for KNN), which is the species pair our RQ needs to explain."*

1.4. Feature Selection — 0.5 / 0.5 (Excellent)

What earned marks. One embedded and one filter method, both computed on training rows only, with top-3 lists traceable to Table 7. The group noticed that the depth-2 tree gives 0.000 to three features and explained transparently how third place was derived.

What held it back. Nothing for this criterion. Note, though, that the observation "the tuned tree assigns 0.000 to petal_length" is the key evidence the Feature Selection Interpretation section later fails to use.

1.5. PCA — 0.5 / 0.5 (Excellent)

What earned marks. Applied to the clustering feature set (Set A), explained variance reported for three components (0.730, 0.229, 0.036), and the top 3 loadings by magnitude correctly identified for each.

What held it back. Nothing for this criterion.

1.6. Clustering — 1 / 1.5 (Good)

What earned marks. Three candidate sets are named and compared with numbers (agreement with species, Table 9). K-Means and complete-linkage hierarchical clustering are both applied to the same set with the same k.

What held it back. The final choice is justified generically ("the most complete description of flower shape") and contradicts the group's own comparison: with Set B, 144 of 149 flowers (96.6%) belong to their cluster's majority species, against 125 (83.9%) for the chosen Set A. The k = 3 justification says "the elbow method", but the largest drop in inertia is from k = 1 to k = 2 (596.0 → 221.0), with a much smaller drop to k = 3 (221.0 → 138.8).

What an Excellent response would say here

"The sharpest bend in the elbow curve is at k = 2 (inertia 596.0 → 221.0, then only 138.8 at k = 3), because versicolor and virginica overlap and form one dense cloud. We nevertheless use k = 3 because our RQ asks whether the three species form natural groups; k = 2 would answer a different question." — and either choose Set B, or explain why Set A is kept despite B's higher scores (for example, that clustering only on petal features would make the comparison with species circular).

2. Data Analysis & Interpretation

2.1. Impact of Preprocessing on Results — 1 / 1.5 (Good)

What earned marks. A before/after value is reported for all three steps (Table 2): 150 → 149 rows; best single-feature MI 1.201 → 1.238 bits; KNN (k = 5) CV macro-F1 0.955 → 0.946.

What held it back. The standardisation result goes *against* the report's own justification — scaling made KNN slightly worse — yet the discussion calls it "a very small difference" and then concludes that preprocessing produced "a cleaner and more consistent dataset". An unexpected result is the most valuable thing to explain, and the reason is in the group's own tables.

What an Excellent response would say here

"Standardisation lowered KNN CV macro-F1 from 0.955 to 0.946. Scaling raised sepal_width's spread from 0.436 to 1.000 — more than doubling its weight in the distance — while sepal_width is the measurement least associated with species (NMI 0.193). At the same time, petal_length, petal_width and petal_area are three strongly related petal-size features (r = 0.963 between the first two), so after scaling, petal size is effectively counted three times and sepal_width adds noise at the boundary between versicolor and virginica. The 0.009 drop is smaller than the fold-to-fold standard deviation (about 0.07), so we treat scaling as neutral for KNN on this dataset, not as an improvement."

2.2. Correlation Interpretation — 1.25 / 2.5 (Good)

What earned marks. All 40 values reported, N/A cases explained, non-causal language throughout, and confounding acknowledged.

What held it back. Agreement and divergence are explained with definitions ("Pearson measures linear, Spearman measures monotonic") rather than with the data. The Pearson–Spearman gap for sepal_width–petal_length is put down to outliers "maybe", although the report had already identified the four outliers on sepal_width by id. Most importantly, the report concludes that sepal length and width "vary largely independently" (r = -0.118) and names "growing conditions" as a hypothetical confounder, while the real confounder is the target variable already in the table: within each species the correlation is positive (setosa 0.743, versicolor 0.526, virginica 0.446). The downstream impact is asserted ("we kept sepal_width because it may still contribute") without evidence that it was tested.

What an Excellent response would say here

"The pooled correlation between sepal_length and sepal_width is -0.118, but within each species it is clearly positive (0.743, 0.526 and 0.446). The pooled value is near zero because setosa flowers have short, wide sepals while the other two species have long, narrow ones — species is a confounder that reverses the within-group relationship (Simpson's paradox). MI detects this structure (0.278 bits) where Pearson does not, because the joint distribution is a mixture of three groups rather than a single linear trend. For our RQ, this means sepal_width is weakly related to species on its own (NMI 0.193) but may still help separate species when combined with sepal_length."

2.3. Model and Evaluation Comparison — 1.25 / 2.5 (Good)

What earned marks. Baseline reported on the same metrics with absolute improvements stated (+0.800 and +0.766 macro-F1); bias/variance discussed with the group's own training and CV values; bootstrap CI computed and its choice justified.

What held it back. The confidence intervals are reported but then misread: the report concludes that "KNN is the better model" although the intervals [0.878, 1.000] and [0.823, 1.000] overlap almost entirely. The gap between the models is one flower — each error on a 30-row test set moves macro-F1 by about 0.033. Feature influence and model behaviour are described generically ("KNN is sensitive to noise") without linking them to the actual misclassified flowers (ids 135, 57, 78), and the RQ conclusion ("petal measurements are the most important") is asserted rather than shown.

What an Excellent response would say here

"KNN made one error (id 135) and the tree two (ids 57 and 78), so the 0.034 difference in macro-F1 is a single flower, and the overlapping bootstrap intervals confirm we cannot rank the models on this test set. What the models do show for our RQ is where species are hard to tell apart: the tree's first split (petal_width ≤ 0.80) separates all setosa, and every error lies between versicolor and virginica. KNN and the tree fail on different flowers in that region, which suggests the two models use petal measurements differently rather than one being better overall."

2.4. Feature Selection Interpretation — 1 / 2 (Good)

What earned marks. The disagreement on third place is identified (sepal_length vs petal_length), and a genuine hard case is chosen (id 135) with its attribute values quoted.

What held it back. Two of the three elements are thin. The scenario for opposite rankings says this "could apply to petal features" — but Table 7 already shows it happening: petal_length has a filter score of 1.000 (rank 3) and a tuned-tree importance of 0.000. The hard-case explanation ("near the boundary between versicolor and virginica") would fit almost any misclassified flower and does not use the information available about id 135.

What an Excellent response would say here

Scenario: "petal_length scores 1.000 nats under MI but 0.000 in the tuned tree. MI scores each feature on its own, and petal_length is strongly associated with species. The tree, however, splits first on petal_width (≤ 0.80) and then on petal_area (≤ 7.42); because petal_length is correlated with petal_width at r = 0.963 and is part of petal_area, it leaves no remaining impurity to reduce, so the tree never uses it."

Hard case: "id 135 is a virginica with petal_width 1.4 — lower than every virginica in the training set (minimum 1.5) and the only virginica of 49 with petal_width ≤ 1.4. Its three nearest training neighbours after scaling are ids 84 and 73 (versicolor) and 134 (virginica), so KNN votes 2–1 for versicolor. The tree classifies it correctly only because petal_area = 7.84 exceeds the 7.42 threshold, yet training versicolor flowers reach petal_area 8.64, so a slightly different split point would have misclassified it too."

2.5. Limitations and Improvement Opportunities — 0.5 / 1.5 (Basic)

What earned marks. Every required area has its own subheading, and alternatives are mentioned.

What held it back. Almost every sentence could be copied unchanged into another report: "the dataset is small", "Pearson is sensitive to outliers", "other clustering methods such as DBSCAN could be explored". One statement is incorrect (z-score standardisation does not assume normality). None of the limitations uses a value, a row or a decision from this group's own analysis, and none of the alternatives is discussed.

What an Excellent response would say here

Preprocessing: "Rejecting IQR removal kept ids 16, 33 and 34, all setosa; removing them would have taken 3 of 50 setosa flowers out of the one species that is already perfectly separated, so keeping them had no cost. Standardisation lowered KNN CV macro-F1 (0.955 → 0.946); a feature-weighted distance or KNN without petal_area are untested alternatives."

Supervised learning: "With 30 test flowers each error costs about 0.033 macro-F1, so the 0.034 gap between models is within one error; repeated stratified CV would compare the models on more than one split."

Feature selection: "The top three MI scores (1.023, 1.001, 1.000) differ by less than their variation across random seeds (standard deviation 0.003–0.007 across 10 seeds), so the filter ranking within the petal features is not stable."

Clustering: "Set B would have recovered species better (144 vs 125 of 149 flowers in their cluster's majority species). Complete linkage placed setosa id 42 (sepal_width 2.3, well below the setosa mean of 3.43) in the versicolor-dominated cluster, showing that sepal_width outliers affect hierarchical clustering."

2.6. PCA Interpretation — 1 / 1.5 (Good)

What earned marks. Each component is interpreted from its actual top 3 loadings, with values.

What held it back. The interpretation is mechanical in places: petal_width's loading on PC2 (0.068) is listed as a "top 3" contributor without noting that it is close to zero. The RQ connection is asserted ("supports the finding") rather than argued. The key insight is missed: the component that explains the second-largest share of variance is almost unrelated to species.

What an Excellent response would say here

"PC1 (73.0% of variance) separates species: setosa PC1 scores range from -2.76 to -1.81, while every versicolor and virginica score is above -0.48, and PC1 has an NMI of 0.595 with species. PC2 explains 22.9% of the variance but has an NMI of only 0.065 with species, because it is dominated by sepal_width (loading 0.925), the measurement least associated with species. For our RQ, this shows that high variance does not mean high relevance: almost all of the species information sits on one axis."

2.7. Clustering Interpretation — 1.25 / 2.5 (Good)

What earned marks. Each K-Means cluster is described by its dominant features (mean petal length 1.46, 4.36, 5.51 cm) and its species composition, and sizes are reported for both methods.

What held it back. The comparison verdict — "both methods produced broadly similar groupings" — is contradicted by the sizes in the same table: K-Means gives 50/52/47 while hierarchical clustering gives 49/24/76, and the hierarchical method's largest cluster absorbs 29 versicolor alongside 47 virginica. The report cites the sizes but does not read them. The RQ discussion is generic.

What an Excellent response would say here

"The two methods agree on setosa (50 and 49 flowers), but split the versicolor–virginica cloud very differently. Of K-Means cluster 0 (52 flowers), 29 go to hierarchical cluster 0 and 23 to cluster 2, so the hierarchical method splits that cluster almost in half. Complete linkage merges clusters by their two most distant members, which favours compact groups: it keeps a small, tight versicolor cluster of 24 and places the other 29 versicolor with the virginica. For our RQ, this shows that the versicolor–virginica distinction is not a natural gap in the measurements — it depends on where the method chooses to cut."

2.8. Cross-Section Consistency — 0.5 / 1 (Basic)

What earned marks. Nearly all values agree across sections.

What held it back. The Conclusion reports the Decision Tree's macro-F1 as 0.946, but Table 6 and the Discussion give 0.933; 0.946 is the tree's cross-validation score from Table 4. The conclusion (KNN ahead) does not change, so this is a minor discrepancy rather than a contradiction.

What an Excellent response would say here

Report the test-set value (0.933) consistently wherever the tree's final performance is stated, and label CV scores as CV scores.

3. Report Quality

Visualisations — 1 / 1 (Good). Figures are referenced in the text, captioned below in 9 pt, and plot what the discussion relies on (train vs validation curves, elbow curve, PCA projections with cluster colours matched to species).

Formatting — 1 / 1 (Good). Template followed: Calibri 11 pt, single spacing, normal margins, nine pages including references (within the 10-page limit), table captions above and figure captions below in 9 pt.

4. Overall

The report is complete and accurate, and that is worth marks — most of the Methods marks were earned. The marks lost are in interpretation, and in nearly every case the evidence for an Excellent answer was already in the group's own tables. Four patterns account for most of the gap:

- **A result contradicts the explanation, and the report moves on.** Scaling lowered the KNN score; Set B outperformed the chosen set; the elbow curve bent most sharply at k = 2. These are the findings most worth explaining.
- **A number is reported but not interpreted.** The overlapping confidence intervals, the sizes 30 and 70 in Table 10, and petal_length's 0.000 importance were all computed and then not used.
- **Explanations restate definitions instead of the data.** "Pearson measures linear relationships" is true of every dataset; "the pooled r of -0.118 hides a positive relationship within each species" is true of this one.
- **Limitations are written once for any report.** A specific limitation names a row, a value or a decision from this analysis, and says what alternative would test it.

Under the rubric's overall standard, isolated strong elements cannot lift a report to Excellent when interpretation is consistently surface-level; this report sits in the Good range overall.